

THANH LONG LY

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🌐 Thanh Long Ly

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Education

Aug 2023 – Aug 2027

📖 **National Economics University (NEU)**, Hanoi, Vietnam.

Major: Data Science in Economics and Business.

CGPA: 3.55/4.0

Relevant Coursework: Machine Learning, Econometrics, Discrete Math, Data Structures & Algorithms.

Research Experience

Jan. 2025 – Present

📖 **Research Assistant**, Predictive Models and Machine Learning Research Applications, NEU.

- Bridged academic research and business applications by fine-tuning transformer architectures (BERT/RoBERTa) for domain-specific NLP tasks.
- Hosted local seminars and technical training sessions to foster a collaborative research environment and knowledge sharing.
- Currently preparing a manuscript for submission to an international conference.

Sep. 2024 – Mar. 2025

📖 **Student Researcher**, Scientific Student Research Competition, NEU.

- Co-authored one paper published in *Proceedings of the National Scientific Conference: Vietnam's Economy in 2024 and Prospects for 2025*.
- Implemented **spatial econometric models with panel data** to quantify policy impact, utilizing rigorous statistical tests to validate hypothesis significance.
- Recognized with the university-level **"Student Research Award in 2025"**.

Research Publications

Conference Proceedings

- 1 M. D. Phung, B. D. T. Vu, T. L. Ly, T. H. T. Duong, and K. L. Pham, "The regulatory role of provincial governments in the impact of labor shift on agricultural production efficiency in Vietnam," in *Proceedings of the National Scientific Conference: Vietnam's Economy in 2024 and Prospects for 2025*, Promoting Economic Institutional Reform in the New Context, Hanoi, Vietnam: National Economics University Publishing House, 2025, pp. 488–501.

In Progress

- 1 T. P. Trinh, A. T. Tran, T. L. Ly, A. K. Phan, Q. A. Bui, and N. H. Dang, *Agentic B2B2C AI - how AI has been transforming B2B2C e-commerce in Vietnam*, In Preparation, 2026.

Projects

- Jan. 2026 – Present  **Real-Time Multimodal AI for IELTS Assessment** (Personal Project).
- Developed a low-latency (<500ms) voice agent via **Gemini 2.5 Flash Native** (WebSockets) for natural mock exam turn-taking.
 - Implemented an automated scoring pipeline using **IELTS band descriptors** to evaluate fluency, vocabulary, grammar, and pronunciation.
 - Designed a React/Node.js architecture handling real-time PCM audio streaming and persistent session history.
- Oct. 2025 – Dec. 2025  **Domain-Specific Conversational AI** (Champion, *Master AI, Master Career* Hackathon 2025).
- Architected a **hybrid agentic system** (LangGraph + Retion.ai) with specialized REST tools (ROI calculation) to overcome terminology ambiguity in complex B2B/B2C scenarios.
 - Awarded **1st Place** amongst 20+ teams for delivering a high-precision QA solution with comprehensive scenario designs.
- Oct. 2025 – Nov. 2025  **Time-Series Forecasting for Meteorological Data** (Final Group Project, Machine Learning Course).
- Achieved the **highest score** for rigorous application of ML principles (robust train-dev-test splitting to prevent data leakage) and detailed reporting.
 - Engineered a reusable **sklearn pipeline** for consistent transformation (60+ features) and Optuna-driven architecture search/tuning across 29 models (CatBoost/LightGBM/XGBoost, 2,400+ trials).
 - Reached **RMSE 1.52°C / R² 0.902** (daily) and **RMSE 1.61°C / R² 0.907** (hourly), deploying the solution via FastAPI and ClearML.

Working Experience

- Jun. 2025 – Present  **AI Engineer Intern**, Alpha Data Academy, Hanoi, Vietnam.
- Developed an **Agentic RAG System** for Rang Dong (leading lighting manufacturer), designing a **hybrid retrieval engine** (dense + sparse + RRF reranking) on **Qdrant**.
 - Built robust data ingestion pipelines and optimized LLM responses to align with business intents; deployed on **Render** with **Opik** monitoring and **MongoDB** logging.

Skills

- Research Stack  **Languages:** Python, C++, SQL.
ML/AI: PyTorch, TensorFlow, HuggingFace Transformers, LangChain, Scikit-learn.
- Engineering  **Backend/DB:** Qdrant (Vector DB), PostgreSQL, MongoDB, Supabase.
DevOps/Tools: Docker, Kubernetes, AWS, Git, MLFlow.
- Languages  English, Vietnamese, Chinese.